

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _Sowmiya . D (192511063) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

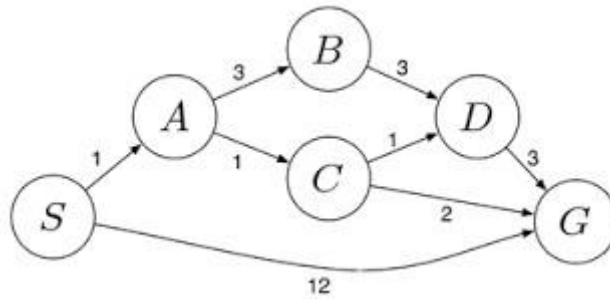
Signature of the student: Sowmiya. D

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem

Given:

- One **4-gallon jug**
- One **3-gallon jug**
- No measuring marks on either jug
- A water pump is available
- Water can be poured onto the ground or between jugs

Goal:

To obtain exactly **2 gallons of water in the 4-gallon jug**.

State Representation:

Let the state be represented as **(x, y)**, where:

- **x** = amount of water in the 4-gallon jug
- **y** = amount of water in the 3-gallon jug

Initial state: **(0, 0)**

Goal state: **(2, 0)** or any state where the 4-gallon jug contains 2 gallons.

Step	Action	State (4-gallon, 3-gallon)
1	Start with both jugs empty	(0, 0)
2	Fill the 3-gallon jug	(0, 3)
3	Pour the 3-gallon jug into the 4-gallon jug	(3, 0)
4	Fill the 3-gallon jug again	(3, 3)
5	Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full	(4, 2)
6	Empty the 4-gallon jug onto the ground	(0, 2)
7	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	(2, 0)

2. Mars Rover as an Intelligent Agent

A Mars Rover is an intelligent agent that autonomously explores the surface of Mars, collects samples, analyzes them, and sends scientific information back to Earth.

i. Percepts of the Agent

The rover perceives its environment using sensors and scientific instruments. Its percepts include:

- Images and videos of the Martian surface
- Distance from rocks and obstacles

- Soil and rock composition
- Temperature, pressure, and atmospheric conditions
- Presence of water or ice
- Current location and position
- Battery level and system status

ii. Operating Environment

The operating environment of a Mars Rover can be characterized as follows:

- **Partially observable:** The complete environment cannot be observed at once.
- **Single-agent:** The rover operates independently on Mars.
- **Stochastic:** The results of actions may be uncertain because of unknown terrain and conditions.
- **Sequential:** Present actions affect future decisions and mission progress.
- **Dynamic:** Weather, temperature, and environmental conditions can change.
- **Continuous:** Movement, time, distance, and measurements are continuous.
- **Unknown:** The rover initially has incomplete knowledge of its surroundings.

iii. Actions the Agent Can Take

The Mars Rover can perform the following actions:

- Move forward, backward, and turn
- Navigate to a selected location
- Detect and avoid obstacles
- Capture images and videos
- Collect soil and rock samples
- Analyze samples using scientific instruments
- Measure environmental conditions
- Send data to Earth
- Manage its battery and energy consumption
- Make autonomous decisions based on sensor data

iv. Evaluation of the Agent's Performance

The rover's performance can be evaluated based on:

- Quantity and quality of scientific data collected
- Number of samples successfully collected and analyzed

- Accuracy of navigation
- Successful obstacle avoidance
- Distance covered during the mission
- Energy and battery efficiency
- Amount of useful information sent to Earth
- Achievement of mission objectives
- Safety and reliability of the rover

v. Suitable Agent Architecture

A **hybrid agent architecture combining model-based, goal-based, and utility-based agents** is most suitable for a Mars Rover.

- A **model-based agent** maintains an internal model of the Martian environment because the rover cannot observe everything directly.
- A **goal-based agent** helps the rover achieve objectives such as reaching a destination, collecting samples, and analyzing them.
- A **utility-based agent** helps the rover choose the safest and most efficient action by considering battery power, distance, scientific value, and risks.

Conclusion: Therefore, a hybrid intelligent agent is most suitable for a Mars Rover because it can perceive its environment, maintain internal knowledge, achieve mission goals, and make efficient decisions autonomously under uncertain conditions.

3. 8-Queens Problem — Search Space and Problem Formulation

Problem

Place **8 queens on an 8×8 chessboard** such that no two queens attack each other.

A queen can attack another queen if they are in the same:

- Row
- Column
- Diagonal

Problem Formulation

Initial State

An empty chessboard with no queens placed.

State

A configuration of queens placed on the chessboard.

A convenient representation is:

[Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8]

where each value represents the row position of the queen in the corresponding column.

Example:

[1, 5, 8, 6, 3, 7, 2, 4]

means one queen is placed in each column.

Actions

Place a queen in an empty column and row where it does not conflict with already placed queens.

Transition Model

The action changes the board by adding one queen to an empty position.

Goal Test

The goal is reached when:

- 8 queens are placed.
- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Path Cost

Each queen placement can be considered as having a cost of 1. The total cost is the number of queen placements.

Search Space

The search space contains all possible configurations of placing 8 queens on the chessboard. If one queen is placed in each column, there are initially:

$$8^8 = 16,777,216$$

possible arrangements before eliminating invalid states.

Using the constraint that no two queens can occupy the same row, the possibilities are reduced to:

$$8! = 40,320$$

arrangements.

Search strategies such as **Depth-First Search, Backtracking, Breadth-First Search, or Constraint Satisfaction Search** can be used to find a solution.

Conclusion: The 8-Queens problem is a constraint-based search problem where the solution is a board configuration satisfying all queen attack constraints.

4. OLA Cab Booking as an Intelligent Agent — Problem-Solving Agent

Problem

A customer wants to travel from one location to another using an OLA Cab booking application. The customer can select different cab types such as **Mini, Micro, Sedan, Shared, and Prime** according to comfort and requirements.

Type of Agent

The most suitable agent is a **Goal-Based Agent**.

It selects an appropriate cab and plans a route to achieve the goal of taking the customer from the source location to the destination.

It can also be considered a **Utility-Based Agent** because it can select the best cab based on:

- Cost
- Comfort
- Travel time
- Distance
- Availability
- Customer preference

Problem Formulation

Initial State

Customer is at the starting location.

Goal State

Customer reaches the selected destination.

Actions

- Enter pickup location
- Enter destination
- Select cab type
- Check cab availability
- Book a cab
- Assign a driver
- Travel to the destination

State

The current location of the customer, available cabs, selected cab type, and destination.

Goal Test

Check whether the customer has reached the destination.

Path Cost

The cost may include:

- Fare
- Travel time
- Distance
- Comfort level

Pseudocode

START

Input pickup_location

Input destination

Display available cab types:

Mini, Micro, Sedan, Shared, Prime

Input customer preference

Search for available cabs

IF cab is available THEN

 Select the suitable cab

 Calculate fare and estimated time

 Confirm booking

 Assign driver

 Start journey

 WHILE current_location != destination

 Track vehicle location

 Navigate towards destination

 END WHILE

 Display "Destination Reached"

ELSE

 Display "No Cab Available"

END IF

STOP

Conclusion: A **goal-based agent** is suitable for reaching the destination, while a **utility-based agent** can improve the decision by selecting the most suitable cab based on cost, comfort, time, and customer preferences.

5. Uniform Cost Search for Logistics Delivery Network (10 Marks)

Problem

A logistics company wants to find the **least-cost path from warehouse S to warehouse**

G. Each route has a cost representing travel expenses such as fuel and time.

The graph is weighted, so **Uniform Cost Search (UCS)** is used.

Given Graph

The important paths and costs are:

- **S → A = 1**
- **A → B = 3**
- **A → C = 1**
- **B → D = 3**
- **C → D = 1**
- **C → G = 2**

- $D \rightarrow G = 3$
- $S \rightarrow G = 12$

UCS Principle

Uniform Cost Search always expands the node with the **lowest cumulative path cost $g(n)$** .

Step-by-Step Search

Step	Expanded Node	Path	Cost
1	S	S	0
2	A	$S \rightarrow A$	1
3	C	$S \rightarrow A \rightarrow C$	2
4	D	$S \rightarrow A \rightarrow C \rightarrow D$	3
5	G	$S \rightarrow A \rightarrow C \rightarrow G$	4

Other possible paths:

- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$
Cost = $1 + 3 + 3 + 3 = 10$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$
Cost = $1 + 1 + 1 + 3 = 6$
- $S \rightarrow A \rightarrow C \rightarrow G$
Cost = $1 + 1 + 2 = 4$
- $S \rightarrow G$
Cost = **12**

Final Answer

The least-cost path found by Uniform Cost Search is:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost:

$1 + 1 + 2 = 4$

Therefore, the **minimum-cost delivery route** from warehouse S to warehouse G is $S \rightarrow A \rightarrow C \rightarrow G$ with a total cost of 4.

UCS guarantees an optimal solution when all path costs are non-negative